

Eklakh Dewan

Software Engineer | AI/ML Engineer | Generative AI

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LinkedIn | GitHub | Portfolio | LeetCode

Professional Summary

B.Tech Artificial Intelligence & Data Science student with hands-on experience building software systems, machine learning applications, backend APIs, and Generative AI solutions. Strong foundation in Data Structures & Algorithms, Object-Oriented Programming, DBMS, SQL, and software engineering, with practical experience in Python, Java, FastAPI, MySQL, Git, and Docker. Built end-to-end systems spanning machine learning, NLP, recommendation systems, RAG, information retrieval, AI workflow automation, and backend engineering. Interested in building reliable, scalable, and intelligent systems that solve practical problems.

Technical Skills

Programming Languages: Python, Java, C++, JavaScript, SQL

Computer Science: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Software Engineering, Complexity Analysis

Software Engineering & Backend: FastAPI, Spring Boot, REST APIs, API Design, Backend Development, JDBC, JSON, Request Validation, Exception Handling, Modular Architecture, Debugging

AI & Machine Learning: Machine Learning, Scikit-learn, TensorFlow, Keras, XGBoost, Classification, Regression, Feature Engineering, Feature Selection, Hyperparameter Tuning, Cross-Validation, Model Evaluation, SHAP

Natural Language Processing: NLP, Transformers, Hugging Face, TF-IDF, Semantic Embeddings, Text Preprocessing, Tokenization, Intent Classification, Named Entity Recognition, Multilingual NLP

Generative AI & Retrieval: RAG, LLM Applications, Prompt Engineering, Embeddings, Semantic Search, Information Retrieval, BM25, FAISS, Dense Retrieval, Hybrid Search, Cross-Encoder Reranking, Citation Validation, Faithfulness Evaluation

Data & Analytics: Pandas, NumPy, ETL Pipelines, Data Preprocessing, Exploratory Data Analysis, Statistical Analysis, Data Visualization, Tableau, Plotly, Matplotlib, Streamlit

Databases: MySQL, MongoDB, SQL Joins, Indexing, Normalization, Database Design, Query Optimization, CRUD

Tools & Platforms: Git, GitHub, Docker, Linux, Postman, Jupyter, AWS Fundamentals, AWS Certified Cloud Practitioner

Experience

AI/ML Engineering Intern

Jan 2025 – Feb 2025

Flowrage Technology | Remote

- Developed Python-based REST APIs using FastAPI to integrate machine learning models with backend services and automated prediction workflows.
- Built data preprocessing pipelines for structured and multilingual text data, including cleaning, normalization, transformation, and feature preparation.
- Performed feature engineering and model optimization using statistical analysis and Scikit-learn, contributing to an approximately 12% improvement in prediction accuracy.
- Optimized Python-based inference workflows, reducing model serving latency by approximately 20%.
- Optimized SQL workflows and database indexing strategies to improve data access efficiency and backend performance.
- Implemented structured logging, validation, and exception handling to improve debugging, reliability, and maintainability of ML and API workflows.
- Collaborated through Git-based development workflows, feature branches, peer code reviews, debugging sessions, and modular software development practices.

Selected Projects

Enterprise RAG & AI Search Platform

2025

Python, FastAPI, FAISS, BM25, SentenceTransformers, React

- Engineered an evidence-grounded Retrieval-Augmented Generation system covering document ingestion, parsing, chunking, metadata extraction, indexing, retrieval, reranking, and answer generation.
- Combined dense vector retrieval with BM25 sparse retrieval and cross-encoder reranking to improve evidence selection and retrieval quality.
- Built modular pipelines for document processing, query processing, evidence retrieval, citation mapping, and evaluation.
- Designed quantitative evaluation using Recall@5, MRR, nDCG, citation validity, citation accuracy, faithfulness, la-

tency, and token-cost measurements.

AI Agent Workflow Automation Platform

2025

Python, LangChain, FastAPI, LLM APIs, YAML, Docker

- Designed an LLM-powered workflow orchestration platform for multi-step task execution and decision support using structured prompts and modular API integrations.
- Built a YAML-driven configuration layer that converts declarative workflow definitions into reusable executable workflows without modifying core application logic.
- Implemented modular agent pipelines with API interactions, conditional workflows, and structured task execution.
- Containerized the platform using Docker for reproducible development and deployment environments.

AI Career Path Recommendation System

2024

Python, Scikit-learn, SHAP, FastAPI, Streamlit

- Built an explainable recommendation system that generates personalized career recommendations from structured skill profiles.
- Applied feature engineering, classification/ranking methods, and model evaluation to identify relevant career pathways.
- Integrated SHAP-based explanations to expose feature-level reasoning behind individual recommendations.
- Developed a FastAPI backend and Streamlit interface for real-time recommendation delivery and interactive analysis.

Job Recommendation & Resume Matching System

2024

Python, NLP, BERT, TF-IDF, Semantic Embeddings, SQL

- Developed a resume-to-job matching system using NLP, TF-IDF, semantic embeddings, cosine similarity, and relevance ranking.
- Extracted structured skills and requirements from resumes and job descriptions to identify candidate-job alignment and skill gaps.
- Designed a normalized SQL schema with indexed relationships and batch similarity computation for efficient job ranking.

Transaction Processing Backend

2024

Java, Spring Boot, MySQL, REST APIs

- Developed a transaction-processing backend using layered Controller–Service–Repository architecture to separate business logic and application responsibilities.
- Implemented REST APIs, input validation, exception handling, transaction management, and MySQL persistence.
- Applied relational database concepts including normalization, SQL joins, indexing, CRUD operations, and transactional workflows.

Multilingual AI Chatbot

2024

Python, Hugging Face Transformers, NLP, FastAPI

- Built a multilingual NLP pipeline supporting 3+ languages through language detection, tokenization, normalization, and language-specific preprocessing.
- Implemented intent classification and named entity recognition using transformer-based models.
- Exposed the NLP pipeline through REST APIs for scalable application integration.

Research & Technical Exploration

HCAD-RAG Research Specification & Literature Synthesis

- Conducted literature review across 15–20 research papers covering Retrieval-Augmented Generation, information retrieval, evidence attribution, and reliable AI systems.
- Synthesized research findings into a reproducible technical roadmap covering document ingestion, hybrid retrieval, ranking, citation tracing, and evaluation.

Enterprise AI Search Research

- Investigated retrieval quality and evidence-grounding techniques through experiments involving dense retrieval, BM25, hybrid search, reranking, and quantitative evaluation.

Education

KPR Institute of Engineering and Technology (KPRIET)

Coimbatore, India

B.Tech in Artificial Intelligence & Data Science

2023 – 2027

CGPA: 8.55/10

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Natural Language Processing, Deep Learning, Generative AI, Software Engineering, Statistics & Probability

Certifications

- AWS Certified Cloud Practitioner – Amazon Web Services
- Natural Language Processing (Elite) – NPTEL
- Business Intelligence & Analytics (Elite) – NPTEL
- Deep Learning Specialization – Coursera / DeepLearning.AI
- Google Data Analytics Professional Certificate – Coursera

Achievements & Problem Solving

- Solved 250+ coding problems across LeetCode and GeeksforGeeks covering arrays, strings, hashing, trees, graphs, recursion, dynamic programming, sorting, and searching.
- Improved ML model prediction accuracy by approximately 12% through feature engineering and model optimization during industry internship.
- Reduced ML inference latency by approximately 20% through optimization of Python-based prediction workflows.
- Built multiple end-to-end applications spanning software engineering, machine learning, NLP, recommendation systems, Generative AI, and information retrieval.

Additional Information

Languages: English, Hindi, Nepali

Professional Interests: Artificial Intelligence, Generative AI, Machine Learning, Natural Language Processing, Information Retrieval, Backend Engineering, Data Systems, Responsible AI